

Project Viva Voce

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Attention-Enhanced U-Net with Transformer Encoder for Fetal Head Segmentation and Growth Parameter Estimation

by

S Sarveshwar (RA2211052010001)

Syed Muzakkir Ahmed (RA2211052010019)

Triloak Ram (RA2211052010034)

Supervisor Name & Designation

Dr. Manikandan A

Assistant Professor, Department of ECE

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Introduction

- Fetal biometric measurements from ultrasound images play a crucial role in monitoring fetal growth and estimating gestational age.
- Among these measurements, Head Circumference (HC), Biparietal Diameter (BPD), and Occipitofrontal Diameter (OFD) are essential indicators of fetal development.
- Traditionally, these measurements are manually obtained by clinicians, which can be time-consuming and subject to inter-observer variability.
- With advancements in image processing and computer vision, automated segmentation and measurement techniques can improve accuracy, consistency, and efficiency.

Literature Survey



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S.No	Title	Authors/Journal	Year	Methodology
1	Deep Learning Techniques for Fetal Ultrasound Image Analysis: A Review	Mushtaq et al. Discover Applied Sciences (Springer)	2025	Reviewed deep learning methods for fetal ultrasound imaging, focusing on fetal head segmentation and key challenges such as noise and unclear boundaries.
2	Automated Fetal Head Segmentation and Head Circumference Measurement Using Deep Learning	Hekal et al. Biomedical Signal Processing and Control (Elsevier / ScienceDirect)	2025	Proposed an automated framework combining deep learning-based segmentation with ellipse fitting to accurately estimate fetal head circumference from ultrasound images.
3	U-Net-Based Framework for Accurate Fetal Head Segmentation in Ultrasound Images	Nagabotu et al. ETRI Journal (Wiley)	2024	Developed an improved U-Net-based segmentation model using hybrid loss functions to enhance fetal head boundary detection in noisy ultrasound images.

S.No	Title	Authors/ Journal	Year	Methodology
4	Advances in Deep Learning for Medical Ultrasound Image Segmentation	Zhang et al. Artificial Intelligence in Medicine (Elsevier)	2023	Reviewed recent deep learning architectures for ultrasound image segmentation, emphasizing encoder–decoder models and attention mechanisms for improved medical image analysis.
5	Attention-Based Deep Learning Models for Medical Image Segmentation	Li et al. IEEE Transactions on Medical Imaging (IEEE)	2022	Investigated attention-enhanced segmentation models for medical imaging applications, demonstrating improved feature localization and robustness in low-contrast ultrasound images.



Problem Statement

- Accurate measurement of fetal head circumference is essential for assessing **gestational age and fetal growth**.
- Manual fetal head segmentation from ultrasound images is **time-consuming** and highly **operator-dependent**.
- Conventional image processing techniques fail to handle **complex anatomical variations** in fetal head shapes.
- Existing automated methods show **limited accuracy and robustness** under varying image quality.
- There is a need for an **automated, reliable, and accurate segmentation system** for fetal head analysis in ultrasound images.

Objective



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- To develop an **automated fetal head segmentation system** using deep learning techniques.
- To accurately extract the **fetal head region** from ultrasound images.
- To enhance segmentation performance using an **U-Net with transformer encoder**.
- To estimate **biometric parameters** reliably from the segmented head region.
- To estimate gestation age and detect abnormality in the ultrasound image.
- To reduce **manual effort and operator dependency** in prenatal ultrasound analysis.
- To support **consistent and accurate fetal growth assessment** in clinical practice.

Relevant SDGs Addressed

- SDG 3:** Good Health and Well-Being
- SDG 9:** Industry, Innovation and Infrastructure
- SDG 10:** Reduced Inequalities



Fig.1: SDG Goals

Block Diagram



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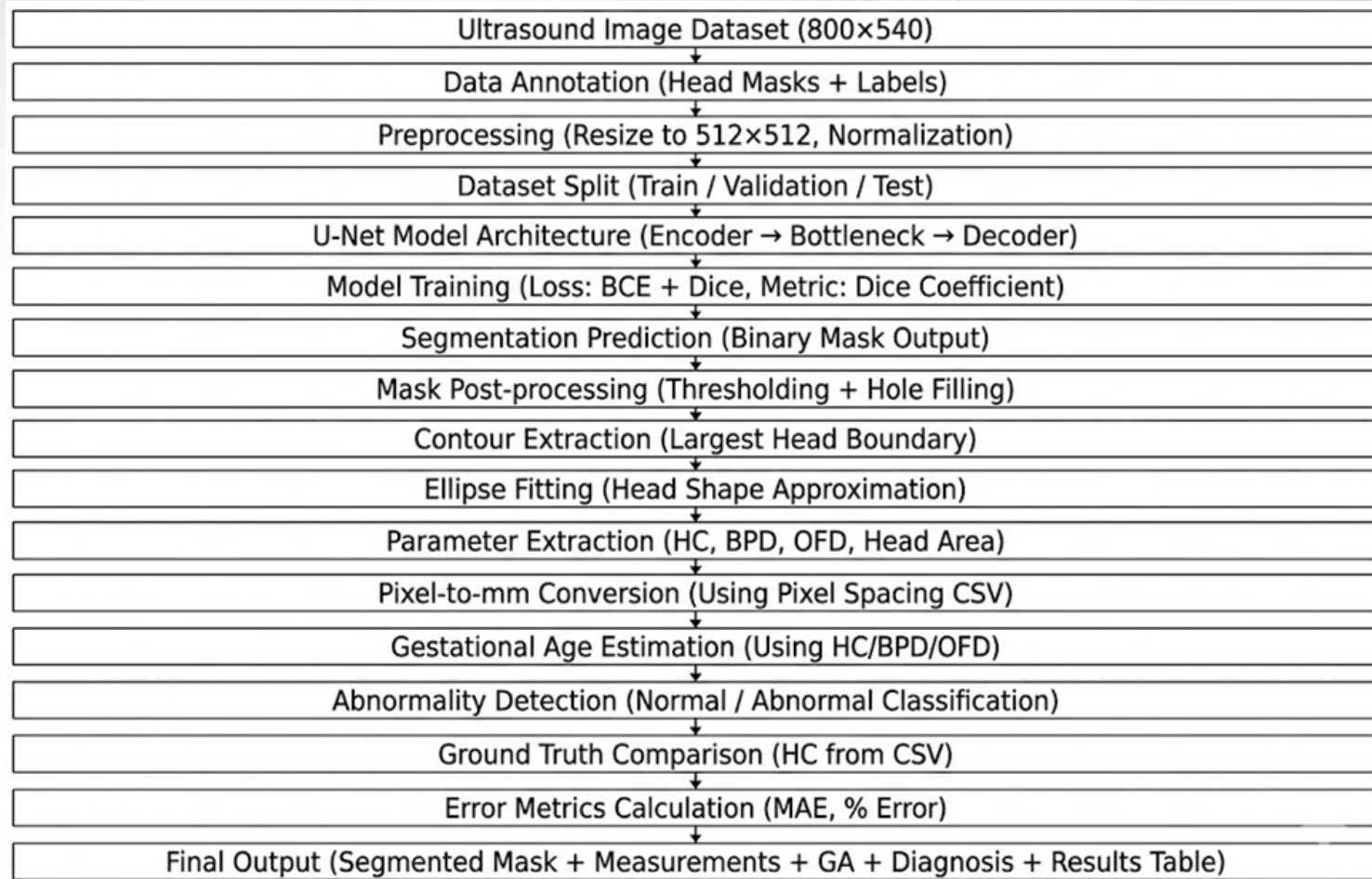


Fig.2: Block Diagram



U-Net Architecture

- The U-Net model is a convolutional neural network designed for biomedical image segmentation. It consists of an encoder-decoder structure with skip connections, which helps in preserving spatial information during the upsampling process.
- U-Net features a symmetric encoder-decoder structure that captures context through downsampling and enables precise localization through upsampling.
- The Transformer Encoder uses Self-Attention to look at the entire image at once. This allows the model to understand the long-range relationships and spatial layout of anatomical structures.
- These connections between encoder and decoder layers help retain spatial information lost during downsampling, improving segmentation accuracy.
- The model produces segmentation masks, binary or multi-class, allowing for pixel-wise classification of input images.
- U-Net performs well even with limited training data.

U-Net Architecture

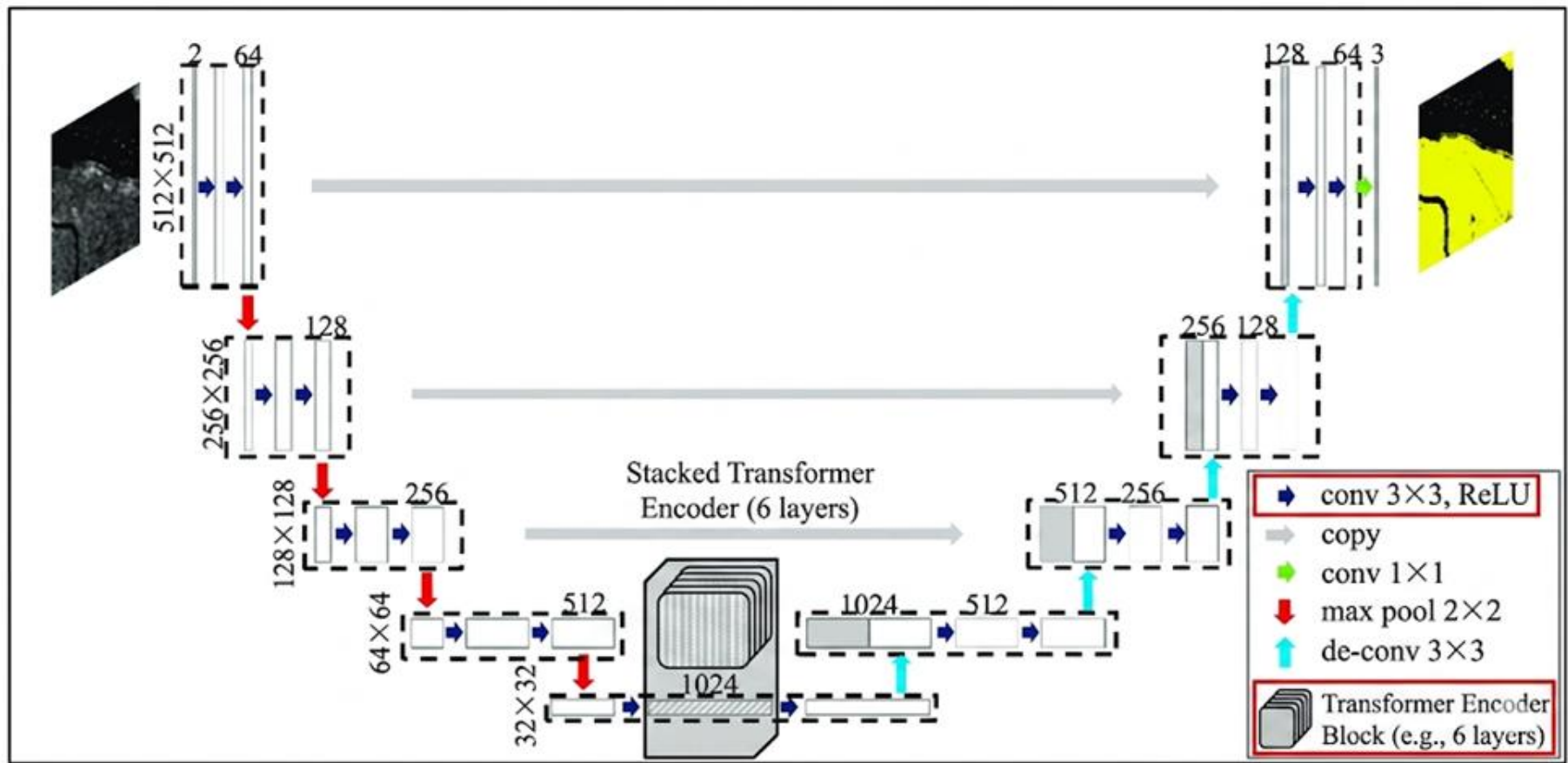


Fig.3: U-Net Architecture



ResNet50 Architecture

- The final diagnostic phase utilizes a customized 50-layer Residual Network (ResNet-50) to learn the subtle morphological differences between normal and abnormal fetal anatomy.
- The model processes the segmented head through four distinct residual groups.
- The network progressively shrinks the spatial dimensions while exponentially increasing the depth of feature maps, capturing highly abstract structural patterns.
- Instead of traditional flattening, the network uses Global Average Pooling (GAP) before its final fully connected layer. This drastically reduces the number of parameters, prevents model overfitting, and outputs a highly reliable binary prediction (Normal vs. Abnormal).

ResNet50 Architecture

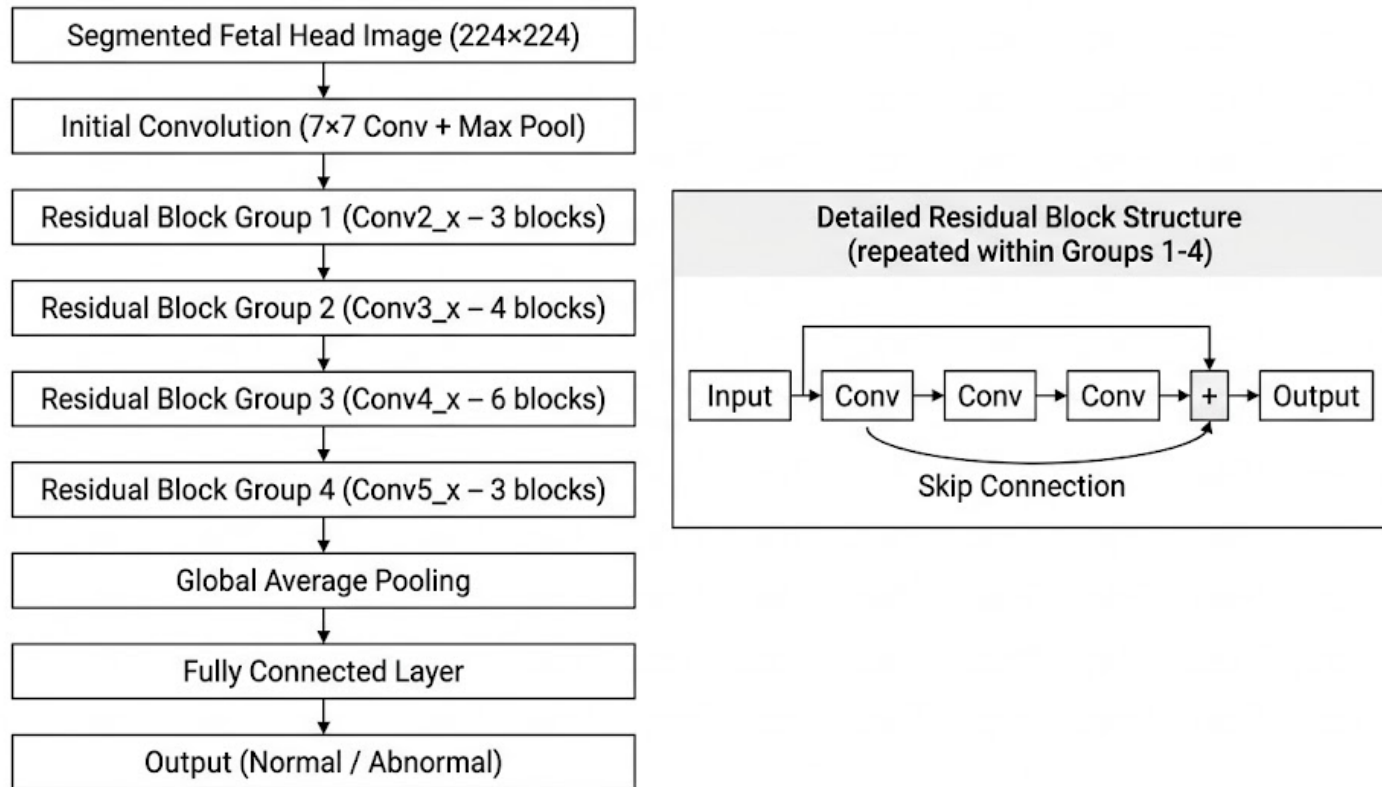


Fig.4: ResNet50 Architecture

Fetal Head Biometric Parameters

1. Head Circumference (HC)

- Total perimeter of the fetal skull.
- Calculated from fitted ellipse or contour.

2. Biparietal Diameter (BPD)

- Maximum side-to-side width of the head.
- Corresponds to the minor axis of the fitted ellipse.

3. Occipitofrontal Diameter (OFD)

- Maximum front-to-back length of the head.
- Corresponds to the major axis of fitted ellipse.
- Used along with to compute HC.

4. Head Area

- 2D cross-sectional area inside skull boundary.
- Calculated using ellipse area formula ($\pi \times \text{semi-major axis} \times \text{semi-minor axis}$).
- Indicates overall head size.

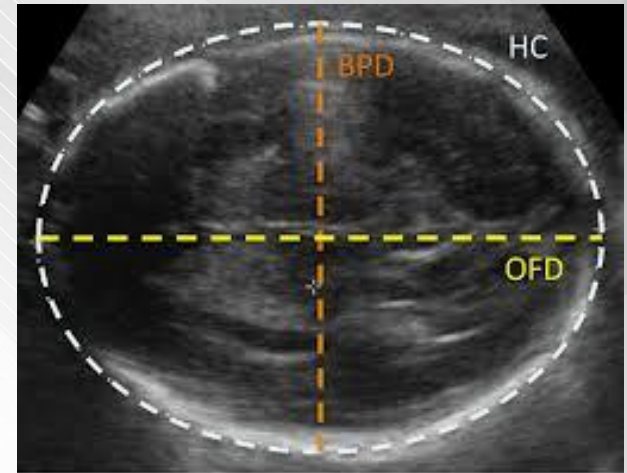


Fig.5: Parameters



Fetal Head Biometric Parameters

$$\text{HC} \approx \pi \left[3(a + b) - \sqrt{(3a + b)(a + 3b)} \right] \quad (1)$$

$$\text{BPD} = 2b \quad (2)$$

$$\text{OFD} = 2a \quad (3)$$

$$\text{Head Area} = N_{\text{pixels}} \times (\text{pixel_size})^2 \quad (4)$$

$$\text{Measurement}_{\text{mm}} = \text{Measurement}_{\text{pixels}} \times \text{pixel_size} \quad (5)$$

where

a = semi-major axis,

b = semi-minor axis,

N_{pixels} = segmented area in pixels.



Gestational Age Estimation

- The predicted Head Circumference (HC) is compared with standard fetal growth charts to estimate the gestational age of the fetus.
- Additional measurements like Biparietal Diameter (BPD) and Occipitofrontal Diameter (OFD) are used alongside HC to improve the accuracy of gestational age estimation.
- Each biometric value is mapped to a corresponding gestational age based on statistical growth curves.
- Combining HC, BPD, and OFD improves estimation reliability and reduces error compared to using a single parameter



Abnormality Detection

- A deep learning model based on ResNet50 is used to classify fetal head images as normal or abnormal.
- The input is the segmented fetal head region, which removes background noise and improves detection accuracy.
- The network extracts high-level features such as shape, texture, and structural patterns of the fetal skull.
- Residual connections help preserve important features and enable training of a deep 50-layer network effectively.
- The final layer outputs a probability score, which is used to determine whether the fetus is normal or shows abnormalities.

Dataset Description

- Images are acquired from the HC18 database (<https://hc18.grand-challenge.org/>).
- There are a total of 806 images with masks, pixel size and head circumference for all images.
- 644 images for training, 42 for testing, and 120 for validation.

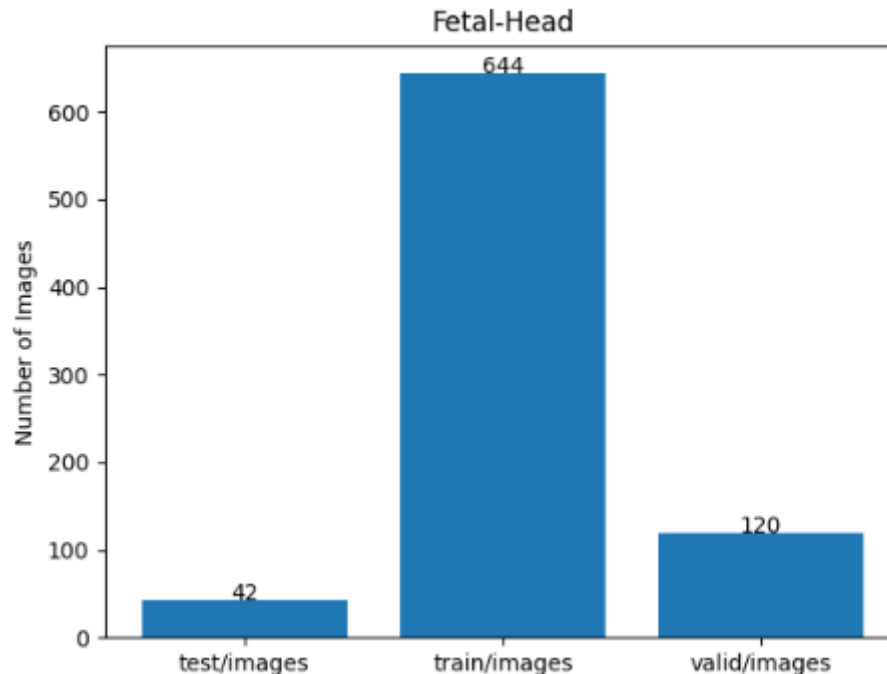


Fig.6: Dataset Distribution



Software Requirements

Programming Language: **Python**

Deep Learning Framework: **TensorFlow**

Image Processing Libraries: **NumPy**

Visualization Tools: **Matplotlib**

Development Environment: **VS Code**

Operating System: **Windows**

Hardware Requirements

Processor: **Intel i5 or higher**

RAM: **8 GB or more**

GPU: **NVIDIA GPU with CUDA support** (for model training)

Storage: **Minimum 20 GB free space**

Segmentation Results

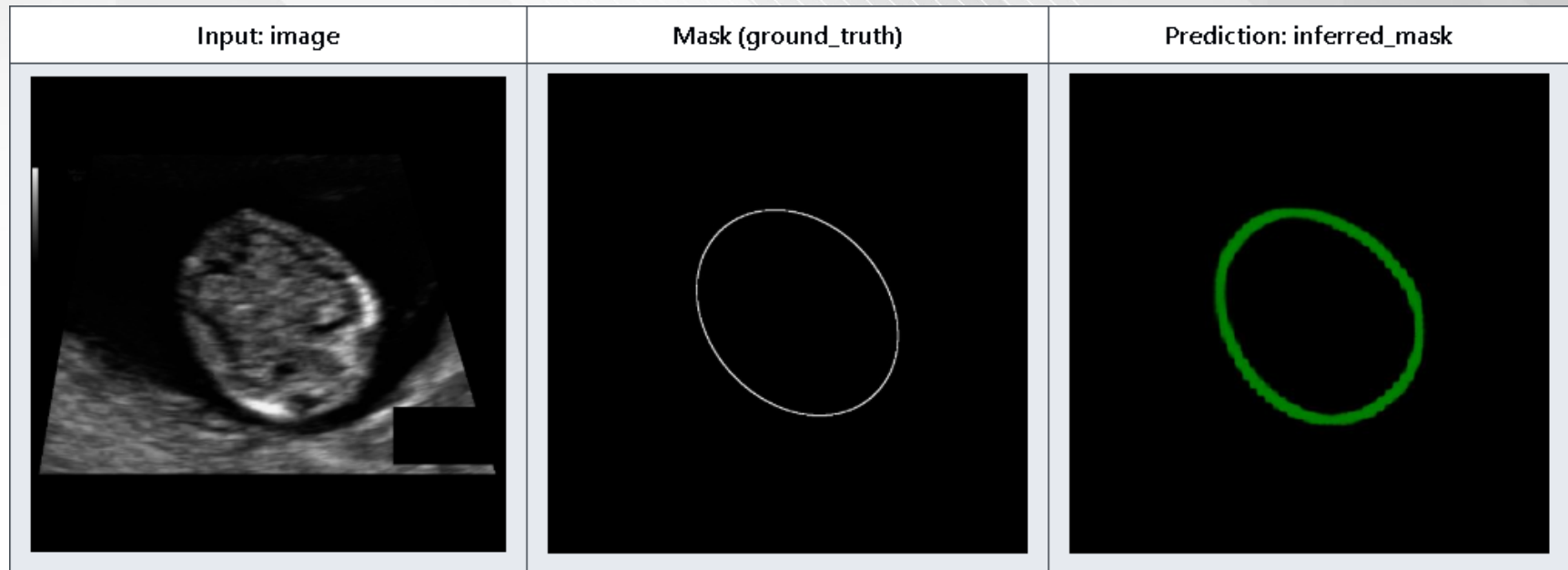


Fig.7: Segmentation result

- **Figure 7** shows improved fetal head segmentation with clearer boundary localization and better agreement with the ground truth.
- The **attention-enhanced U-Net with transformer encoder** improves feature representation in noisy ultrasound images.
- The results indicate **enhancement in segmentation quality** and more reliable head circumference estimation.

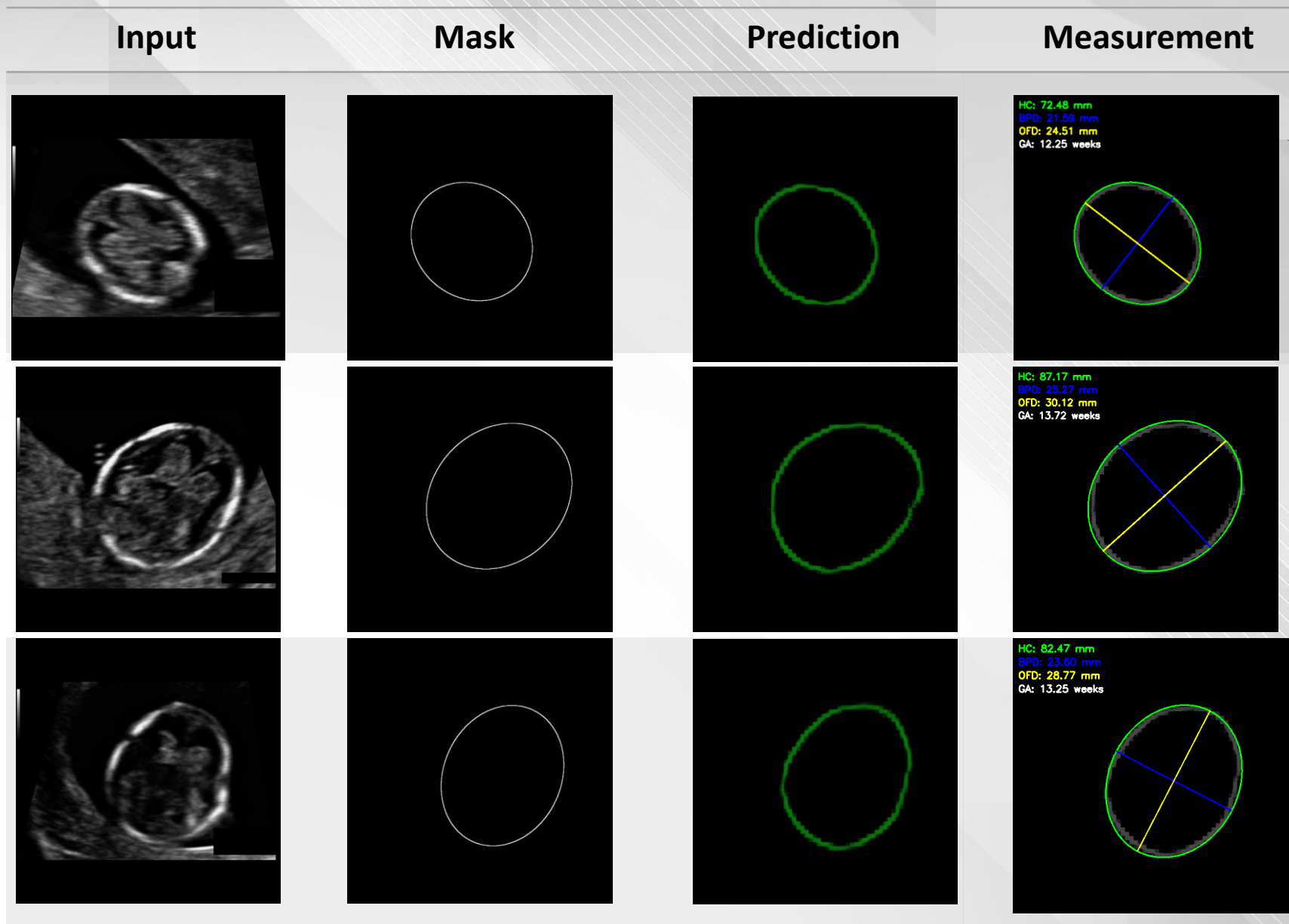


Fig.8: Final result

Results

filename	HC_pred_mm	HC_csv_mm	HC_error_mm	HC_abs_error_mm	HC_error_percent	BPD_mm	OFD_mm	HeadArea_mm2
036_HC.png	74.97223319	73.96	1.012233187	1.012233187	1.368622481	20.82121949	26.72418979	432.9310909
043_HC.png	72.48401581	69.94	2.54401581	2.54401581	3.637426094	21.5852316	24.51302675	412.5336948
050_HC.png	71.21723527	67.5	3.717235275	3.717235275	5.507015222	20.46069768	24.77469007	395.7620999
068_HC.png	87.17307205	83.37	3.803072046	3.803072046	4.561679316	25.26704889	30.12259017	593.1669892
078_HC.png	82.47163192	78.08	4.391631925	4.391631925	5.624528592	23.60100808	28.77424217	528.558066

Fig.8: HC, OFD, BPD and Head Area Measurement

- From the ellipse, BPD is calculated as minor axis, OFD as major axis, head area using ellipse formula and head circumference using ellipse approximation.
- The predicted values are compared with ground truth values from the CSV file and compute absolute and percentage errors.

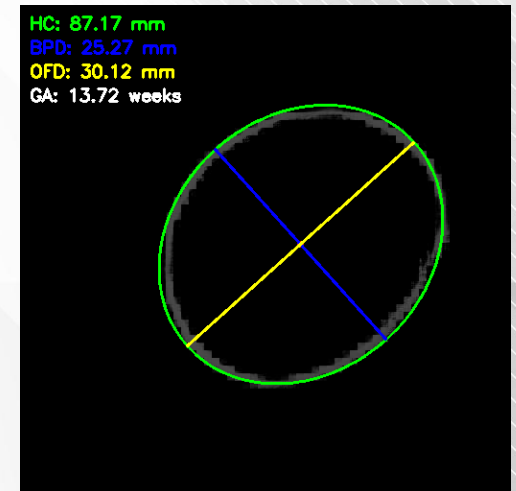


Fig.9: Measurement

Results

```
(tf210) PS U:\project\tensorf\src\abnormal_age> python test_abnormal.py
945_HC.png | True: malignant | Pred: malignant ✓
946_HC.png | True: malignant | Pred: malignant ✓
946_HC_Annotation.png | True: malignant | Pred: malignant ✓
947_HC_Annotation.png | True: malignant | Pred: malignant ✓
949_2HC.png | True: malignant | Pred: malignant ✓
94_HC_Annotation.png | True: malignant | Pred: malignant ✓
958_HC.png | True: malignant | Pred: malignant ✓
961_2HC.png | True: malignant | Pred: malignant ✓
963_4HC.png | True: malignant | Pred: malignant ✓
965_2HC_Annotation.png | True: malignant | Pred: malignant ✓
983_HC.png | True: malignant | Pred: malignant ✓
985_HC_Annotation.png | True: malignant | Pred: malignant ✓
986_HC.png | True: malignant | Pred: malignant ✓
987_2HC_Annotation.png | True: malignant | Pred: malignant ✓
98_HC.png | True: malignant | Pred: malignant ✓

📄 Prediction: non_malignant
(tf210) PS U:\project\tensorf\src\abnormal_age> 
```

Fig.10: Abnormality Detection Result



Conclusion

- U-Net segmentation is combined with geometric ellipse fitting to improve Intersection-over-Union (IoU) scores and predictive precision for fetal cranial sizes.
- By identifying structural boundaries first, we ensure that key measurements (HC, BPD, and OFD) are derived from precise anatomical modeling rather than simple statistical regression.
- The contour-based fitting approach provides high resilience against common imaging noise, such as speckle artifacts and discontinuous edges, maintaining reliable output in clinical settings.
- Eliminates the need for manual measurements, thereby minimizing human error and significantly increasing the efficiency of fetal evaluations.
- Establishes a validated foundation that can be expanded to other fetal biometrics and integrated into real-time clinical decision-support systems.

References

- [1] S. Mushtaq, M. A. Khan, T. Hussain, and S. Kadry, “Deep learning applications in fetal ultrasound imaging: A comprehensive review,” *SN Computer Science*, vol. 6, no. 1, pp. 1–22, 2025.
- [2] M. Hekal, A. Elsharkawy, and A. Elnakib, “Automated fetal head segmentation and head circumference measurement from ultrasound images using deep learning,” *Biomedical Signal Processing and Control*, vol. 89, pp. 105–118, 2025.
- [3] P. Nagabotu, S. Reddy, and R. R. Guddeti, “Accurate fetal head segmentation in ultrasound images using U-Net based deep learning framework,” *Electronics and Telecommunications Research Institute Journal*, vol. 46, no. 2, pp. 1–10, 2024.
- [4] Y. Zhang, L. Wang, and J. Chen, “Advances in deep learning for medical ultrasound image segmentation,” *Artificial Intelligence in Medicine*, vol. 139, pp. 102–118, 2023.
- [5] H. Li, X. Zhou, and Y. Liu, “Attention-based deep learning models for medical image segmentation,” *IEEE Transactions on Medical Imaging*, vol. 41, no. 9, pp. 2456–2468, 2022.



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